

APPENDIX A GLOSSARY

ALGORITHM

A procedure for solving a mathematical problem in a finite number of steps that frequently involves repetition of an operation.

BCD (Binary-Coded-Decimal) Number

The representation of cardinal numbers 0 through 9 by 10 binary codes of any length. The minimum length is four and there are over 29×10^9 possible four-bit BCD codes. However, codes in which the four bits contain the hexadecimal representation of 0 through 9 are the more commonly used codes, for obvious reasons.

BIASED EXPONENT

The sum of the exponent and a constant (bias) chosen to make the biased exponents range non-negative.

BINARY FLOATING-POINT NUMBER

A bit string characterized by three components: a sign, a signed exponent, and a significand. (See single, double, and extended precision.) The numerical value of the bit string is the signed product of the significand and two raised to the power of the exponent.

DENORMALIZED NUMBER

A floating-point number having all zeros in the exponent and a non-zero value in the fraction/mantissa.

DOUBLE PRECISION

A 64-bit binary floating-point operand format composed of three fields: a one-bit sign field, an 11-bit biased exponent field, and an 52-bit fraction (significand) field.

DYADIC OPERATION

An operation on two operands.

E FIELD

See exponent (E field).

EXPONENT

A symbol written above and to the right of a mathematical expression to indicate the operation of rising to a power.

EXPONENT (E FIELD)

The component of a binary floating-point number that normally signifies the integer power to which two is raised in determining the value of the represented number. Occasionally, the exponent is called the signed or unbiased exponent.



ENDED PRECISION

A 96-bit binary floating-point operand format composed of four fields: a one-bit sign field, an 15-bit biased exponent field, a 16-bit undefined field, and a 64-bit mantissa (significand) field.

F FIELD

See fraction (F field).

FIXED-POINT

Pertaining to a numeration system in which the position of the radix point is fixed with respect to one end of the numerals, according to some convention. The integer data types used by the FPCP are fixed-point numbers.

FLOATING-POINT

Pertaining to a system in which the location of the radix point does not remain fixed with respect to one end of the numerical expressions, but is regularly recalculated. The location of the point is usually given by expressing a power of the base (or radix). The single, double, and extended precision data types used by the FPCP are floating-point numbers.

FRACTION (F FIELD)

The field of the significand that lies to the right of its implied binary point.

INTEGER

Any of the natural numbers, the negatives of these numbers, or zero.

MANTISSA

Mantissa and significand are interchangeable throughout this manual. See significand.

MODULO

A mathematical operation that yields the remainder of division. Thus, 39 modulo 6 = 3.

MONADIC OPERATION

An operation on one operand, for example, negation.

NAN (Not-A-Number)

A symbolic entity encoded in floating-point format. There are two types of NANs; signaling and quiet. Signaling NANs signal the valid operation exception whenever appearing as operands. Quiet NANs propagate through almost every arithmetic operation without signaling exceptions.

NORMALIZE

To convert a floating-point number to one whose significand consists of an integer bit of 1.

OPERAND

That which is, or is to be operated upon. An operand is usually identified by an address field of an instruction.

ORTHOGONAL

Statistically independent.

See sign bit (S field).

S FIELD

See sign bit (S field).

SIGN BIT (S FIELD)

Denotes the sign of the operand: zero for positive and one for negative. Floating-point numbers are in sign-magnitude form, which means that only the S bit is complemented to change the sign of the represented number.

SIGNIFICAND

The component of a binary floating-point number that consists of an explicit or implicit leading bit to the left of the implied binary point and a fraction field to the right of the implied binary point. Significand and mantissa are interchangeable throughout this manual. See fraction (F field).

SINGLE PRECISION

A 32-bit binary floating-point operand format composed of three fields: a one-bit sign field, an 8-bit biased exponent field, and a 23-bit fraction (significand) field.

TRANSCENDENTAL

Being, involving, or representing a function (sine x , log x) that cannot be expressed by a finite number of algebraic operations.

UNNORMALIZED NUMBER

An extended precision external operand that contains an explicit integer part bit (i) of zero and an exponent that is neither the maximum nor the minimum for the format.

